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Working English Translation Batch 05
An Algebraic Criterion for Absolute Irreducibility

Translator's note

This working translation begins Noether's paper *Ein algebraisches Kriterium für absolute Irreduzibilität.* I translate the statement of the criterion together with the opening sections, where Noether constructs the reduction polynomials by means of Kronecker substitution. The notation has been lightly regularized, but the algebraic content and the order of exposition are kept close to the source.

An algebraic criterion for absolute irreducibility

A polynomial with coefficients in an arbitrarily given field is called *absolutely irreducible* if it remains irreducible in an algebraically closed field extending the coefficient field. I shall show below that, in contrast with irreducibility relative to a prescribed field, absolute irreducibility can be formulated algebraically. More precisely, the following theorem holds.

Theorem. To every polynomial in $n > 2$ variables with indeterminate coefficients one can attach a whole rational function of those coefficients and of further indeterminates — the *reducibility form* — in such a way that for every specialized polynomial of the same degree the necessary and sufficient condition for absolute irreducibility is the nonvanishing of the reducibility form. In other words: for every specialized system of coefficients for which the reducibility form is not identically zero in the auxiliary indeterminates, the polynomial is absolutely irreducible. Equivalently, the polynomial remains irreducible modulo every prime ideal of every finite algebraic field containing the coefficient field, with at most finitely many exceptional prime ideals. In particular, the containing field may be the rational numbers.

Further applications to the theory of relatively integral functions will be given elsewhere.

§1. The universal product relation

We first show that every polynomial in more than two variables with indeterminate coefficients is absolutely irreducible. Write

$$F(x) = \sum A_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n}, \qquad i_1 + \cdots + i_n = l, \tag{0.1}$$

$$G(x) = \sum B_{j_1 \dots j_n} x_1^{j_1} \cdots x_n^{j_n}, \qquad j_1 + \cdots + j_n = m, \tag{0.2}$$

$$H(x) = F(x)G(x) = \sum C_{k_1 \dots k_n}(A, B) x_1^{k_1} \cdots x_n^{k_n}, \qquad k_1 + \cdots + k_n \leq l + m. \tag{0.3}$$

Here the A and B are indeterminates, and the coefficients $C(A, B)$ are integral bilinear expressions in the A's and B's.

Consequently the quotients

$$D = \frac{C(A, B)}{C_{0 \dots 0}(A, B)}$$

are rational functions of the quotients $A/A_{0\ldots 0}$ and $B/B_{0\ldots 0}$. But the number of such functions is larger than the number of their arguments: namely, the three respective counts are

$$\binom{l+n}{n}-1, \qquad \binom{m+n}{n}-1, \qquad \binom{l+m+n}{n}-1,$$

and one has the inequality

$$\binom{l+n}{n}+\binom{m+n}{n} \geq \frac{(l+m+n)!}{n!(l+m)!} = \binom{l+m+n}{n}.$$

Therefore there is at least one algebraic relation among the $D(A,B)$, and hence also among the coefficients $C(A,B)$:

$$\Phi(Z) \not\equiv 0 \quad \text{(identically in } Z\text{)}, \qquad \Phi(C(A,B)) = 0 \quad \text{(identically in } A, B\text{)}. \tag{0.4}$$

Here Φ denotes an integral polynomial.

Hence a polynomial with indeterminate coefficients,

$$E(x) = \sum Z_{k_1\ldots k_n} x_1^{k_1} \cdots x_n^{k_n}, \qquad k_1 + \cdots + k_n \leq l+m, \tag{0.5}$$

is necessarily absolutely irreducible for $n > 2$.

§2. The prime ideal of reducibility relations

The totality of all such polynomials $\Phi(Z)$ that vanish identically after the substitution $Z = C(A,B)$ forms an ideal in the polynomial ring; more precisely it forms a *prime ideal* $\mathfrak{B}$. Indeed, if a product $\Phi_1\Phi_2$ vanishes identically under the substitution, then at least one factor vanishes identically.

Since relation (0.4) holds identically in A, B, it remains valid for every specialized choice of A, B, where A, B, and therefore also $\Gamma = C(A,B)$, may be elements of an arbitrary field. Thus the coefficients Γ of every polynomial that becomes reducible in an extension field satisfy the relations

$$\Phi(\Gamma) = 0;$$

in other words, they are zeros of the prime ideal $\mathfrak{B}$, equivalently of finitely many basis polynomials for $\mathfrak{B}$. What remains is to prove the converse.

For this it is important to note that all relations among A, B, C remain valid if in the polynomials (0.1)–(0.3) we homogenize by adding a new variable y and pass to homogeneous forms $F(x,y)$, $G(x,y)$, $H(x,y)$ of dimensions l, m, $l+m$. Thus coefficients Γ may occur among the zeros of $\mathfrak{B}$ for which, say, $F(x,y)$ degenerates to a power of y; in passing back to inhomogeneous polynomials this corresponds to a lowering of degree for $H(x)$ rather than to an actual factorization. It will be shown that, apart from this degree-lowering phenomenon, the converse always holds. In particular, for homogeneous forms the converse holds without exception, provided not all coefficients of F vanish; that is, every zero of $\mathfrak{B}$ is produced by a parameter representation

$$\Gamma = C(A,B)$$

with coefficients A, B in a suitable extension field.

Noether proves this by constructing finitely many special functions $\Phi(Z)$ directly. She associates to the polynomials (0.1)–(0.3) corresponding one-variable polynomials by means of Kronecker substitution, shows that gaps occur among the resulting exponents, and from the norms of the corresponding coefficients obtains the desired functions $\Phi(Z)$ and therefore the criterion.

§3. Kronecker substitution

Set

$$d = l + m + 1. \tag{0.6}$$

Then the Kronecker substitution associates to the polynomials F, G, H the one-variable polynomials

$$f(\xi) = \sum A_{i_1 \dots i_n} \, \xi^{i_1 + i_2 d + i_3 d^2 + \cdots + i_n d^{n-1}}, \tag{0.7}$$

$$g(\xi) = \sum B_{j_1 \dots j_n} \, \xi^{j_1 + j_2 d + j_3 d^2 + \cdots + j_n d^{n-1}}, \tag{0.8}$$

$$h(\xi) = \sum C_{k_1 \dots k_n}(A, B) \, \xi^{k_1 + k_2 d + \cdots + k_n d^{n-1}}, \qquad k_1 + \cdots + k_n \leq l + m. \tag{0.9}$$

This correspondence is one-to-one, because from an induced exponent one can recover the exponents in the original monomial: the base-d expansion is unique, and since $d = l + m + 1$ no carrying occurs among the exponents that arise from multiplication.

Thus one has simply

$$h(\xi) = f(\xi) g(\xi), \tag{0.10}$$

and conversely, whenever a relation of the form (0.10) holds in such a way that no exponents other than the induced ones occur in f and g, one may recover the original multivariable factorization

$$H(x) = F(x) G(x). \tag{0.11}$$

Moreover, genuine gaps do appear among the induced exponents in $f(\xi)$ and $g(\xi)$. The highest exponent in $f(\xi)$ is ld^{n-1}, while the second highest is $(l-1)d^{n-1} + d^{n-2}$. Their difference is

$$d^{n-2}(d-1) > 1,$$

because $d = l + m + 1 > 3$ and $n > 2$. The same holds for $g(\xi)$.

The relations (0.10) and (0.11), first established for the indeterminates A, B, remain valid after every specialization of A, B. To preserve the degree one homogenizes (0.10) and (0.11) by means of a new variable η (and likewise y in the multivariable side). Thus a homogeneous form $H(x, y)$ factors in an algebraic extension field if and only if the corresponding binary form $h(\xi, \eta)$ splits into two factors in such a way that only the induced exponents appear in the factors.

This is the point at which Noether passes from the mere existence of algebraic relations to the construction of the reducibility form via norms. The later sections produce the explicit criterion from these one-variable relations.